

MSCC on Raspberry Pi — one how-to

Pi: 4 or 5, **64-bit Raspberry Pi OS** (desktop recommended).

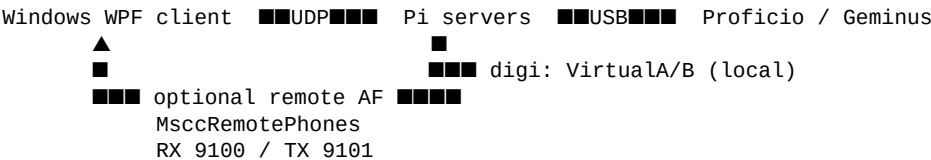
Packages: use files in `pi-install/packages/` (this folder).

Updated: 2026-09 — servers **1.0.40**, init-gui **1.0.13** (hides Proficio/Multus I/Q from speaker/mic pickers), portaudio **19.8.2**, UI **0.6.39**.

If a filename in `packages/` differs, **use the real name on disk**.

Big picture

Role	Where it runs
Servers (ms-sdr, sdrcore-recv, sdrcore-trans)	Always on the Pi (next to the radio)
Operate UI	Windows WPF on a PC, and/or Avalonia MSCC UI on the Pi
Digi apps (WSJT, etc.)	Prefer on the Pi (VirtualA / VirtualB)
Remote operator voice	Optional: phones AF + mic on a Windows PC via MscRemotePhones



Typical setups

- 1. Headless Pi + Windows WPF** — Pi runs servers only; you operate from the PC.
- 2. Pi with Avalonia UI** — servers + **MSCC UI** on the Pi (local `127.0.0.1`).
- 3. Remote voice** — digi on Pi; operator phones/mic on Windows (**Remote Audio**).

What you need

1. Pi on the network; desktop Terminal for install.
2. Contents of `pi-install/packages/` on the Pi (`~/Downloads` or USB).
3. Proficio/Geminus powered when configuring (recommended).
4. Operator audio plan (especially **Pi 4**): prefer **I²S HAT** or digi-only — avoid USB headset + Proficio USB on Pi 4 under load.

A. Install servers (required)

On the Pi Desktop Terminal, `cd` to the folder with the `.deb` files:

```
cd ~/Downloads/packages # or wherever you copied pi-install/packages
```

1) PortAudio (first)

```
sudo apt install -y ./mscc-portaudio_19.8.2_arm64.deb
ldconfig -p | grep portaudio
# expect: /usr/local/lib/libportaudio.so.2
```

2) Main MSCC package (servers + firmware upload tools)

```
sudo apt update
sudo apt install -y ./mscc_1.0.40_arm64.deb
```

Optional helper (if present):

```
chmod +x ../install-mscc.sh # if you copied install-mscc.sh next to packages
../install-mscc.sh
```

This installs:

Piece	Notes		
Servers	\$HOME/mscc/ — ms-sdr, sdrcore-recv, sdrcore-trans		
Firmware Upload	boot loader + boot loader-gui — menu MSCC → Firmware Upload		
Start / Stop / Status	Desktop MSCC menu + `mscc start\`	stop\`	status`
Digi sinks	VirtualA / VirtualB		
Config seed	\$HOME/.local/mscc/ if empty		

3) Setup wizard

```
sudo apt install -y ./mscc-init-gui_1.0.13_all.deb
```

4) Log out / log in once (first install)

Needed so dialout / audio / plugdev groups apply (CAT and devices).

Check digi sinks:

```
pactl list short sinks | grep Virtual
# if empty:
systemctl --user enable --now msc-virtual-audio
```

5) Configure once — MSCC Init

Menu **MSCC** → **MSCC Init** (or msc-init):

- Keyer, CAT / PTT
- Operator speaker & mic (not Proficio I/Q devices)
- Digi fixed to VirtualA / VirtualB
- **Proficio MKII vs Legacy** (PROFICIO-MKII) — use **Legacy (0)** for semi-break-in / external keyer radios
- Optional: start servers at end of wizard

Config lives in \$HOME/.local/mscc/ (upgrades do **not** overwrite).

6) Everyday servers

Menu	Action
MSCC Start	Start servers
MSCC Stop	Stop servers
MSCC Status	Health check
MSCC Init	Reconfigure
Firmware Upload	Flash Proficio/Geminus .cyacd (BOOT jumper → LOADER)

```
mscc start
mscc status
mscc stop
```

B. Install Avalonia UI on the Pi (optional)

Use when you want the operate GUI **on the Pi** (or a second Linux box).

```
sudo apt install -y ./mscc-ui_0.6.39_arm64.deb
```

Item	Detail
Run	Menu MSCC UI , or <code>mscc-ui</code>
Default host	127.0.0.1 port 8888 (start servers first)
Settings	~/.config/MSCC/mscc-avalonia.ini (survive reinstall)
Uninstall	<code>sudo apt remove msc-ui</code> (does not remove servers)

UI features to know (0.6.39+)

Feature	Where / notes
CQ / keyer memory	CW tab — 4 slots, R store / P play (MKII + keyer).
Remote Audio	Left rail checkbox — with Phones selected, sends audio device 2 (remote mic). Greyed on Digital . Sticky in ini.
External electronic keyer (legacy)	CW tab — sets PROFICIO-MKII=0 in host <code>mscc.ini</code> ; restart servers to apply. HOLD stays live.
Spectrum / waterfall	MAIN tab. Heals pan refresh on connect. Packet line shows D5 / Spec. 0.6.39 denser keep-alives + larger UDP buffers so pan flood does not trip "keep-alive lost".
Connect	Host / ports for remote Pi if UI is not on the radio machine

Note: Rebuild the UI .deb after client code changes (Avalonia-Migration/build-mscc-ui-deb.ps1), then refresh pi-install/packages/ with `collect-packages.ps1`.

More detail: `mscc-ui/Avalonia-Migration/INSTALL-MSCC-UI.md`.

C. Windows client (WPF) against the Pi

1. Pi: **MSCC Start**; note Pi IP (`hostname -I`).
2. Windows: run **MSCC.Wpf** from `C:\mscc-net9` (or your install).
3. Settings: Server IP = Pi; **Launch Servers** usually **off** (servers already on Pi).
4. **Start** / Connect.

Legacy CW: check **Use external electronic keyer** on CW tab, or set `PROFICIO-MKII=0` on the Pi (`mscc.ini` / Init / MSCC-Remote if used on Windows host).

D. Remote audio (operator voice on Windows, digi on Pi)

Servers already support it. You need the companion app + UI checkbox.

On the Pi

Edit (or create) phones stream target — Windows PC IP:

```
$HOME/.local/mscc/remote-phones.ini
```

```
ENABLED=1
HOST=192.168.x.x
PORT=9100
```

Mic listen port (usually default is enough):

```
$HOME/.local/mscc/remote-mic.ini
```

```
PORT=9101
```

Restart servers after editing. Digi stays on VirtualA/B.

On Windows

1. Build/run **MscRemotePhones** from `mscc-remote-audio/` (see that folder's README).
2. **RX**: listen **9100** (phones AF from Pi).
3. **TX**: host = **Pi IP**, port **9101**, start mic TX.
4. In **WPF or Avalonia**: Audio = **Phones**, check **Remote Audio** → client sends 0x9B / **data 2**.
5. Digital path: leave **Digital** selected for digi; Remote Audio is ignored (always device 0).

Full handoff: `mscc-remote-audio/STEW-REMOTE-AUDIO.md`.

Test without UI (live session):

```
cd ...\\mscc-remote-audio
.\Set-AudioDevice.ps1 -HostName PI_HOSTNAME -Mode remote
```

E. Firmware upload (bootloader)

Included in the `mscc` package — no separate `.deb`.

1. Power off radio → install **BOOT** jumper → power on → Morse **LOADER** / USB 04b4:b71d.
2. **Stop** MSCC servers.
3. Pi menu **MSCC** → **Firmware Upload** (boot loader-gui) or CLI boot loader `/path/to/file.cyacd`.
4. Use the correct tree's `.cyacd` (Proficio/Geminus MKII or Legacy under Release-*).
5. Power off → **remove BOOT jumper** → power on → Proficio 16c0:05dc.

Details: `Release-Proficio-Legacy/STEW-FIRMWARE-UPDATE.md` (same procedure for MKII trees).

F. Updating packages later

1. On the Windows/dev PC, put new builds in the usual trees (`mscc-deb/`, `mscc-ui/Avalonia-Migration/`, etc.).
2. Refresh the kit:

```
cd C:\Users\n8vet\OneDrive\Documents\GitHub\mscc-station
powershell -NoProfile -ExecutionPolicy Bypass -File .\pi-install\collect-packages.ps1
```

3. Copy new files from `pi-install/packages/` to the Pi.

4. On the Pi:

```
cd ~/Downloads/packages
sudo apt install -y ./mscc-portaudio_....deb ./mscc_....deb ./mscc-init-gui_....deb ./mscc-ui_....deb
```

`apt install ./file.deb` upgrades in place. **Config under** `~/local/mscc/` **and UI** `~/config/MSCC/` **is kept.**

Then:

```
mscc restart
# or MSCC Stop / Start from the menu
```

Update the version table at the top of this file when you change the kit.

Quick reference — where things live in the repo

Need	Path
This kit + how-to	pi-install/
Server packaging / older install prose	mscc-deb/
Init GUI sources	mscc-init-gui/
Avalonia UI + build script	mscc-ui/Avalonia-Migration/
Windows WPF	mscc-ui/windows-work-tree/
Remote phones app	mscc-remote-audio/
Radio .cyacd	Release-Proficio-*, Release-Geminus-*

Checklist (first Pi)

- [] PortAudio → main msc → init-gui → log out/in
- [] VirtualA/B present; **MSCC Init** done (PROFICIO-MKII correct)
- [] **MSCC Start**; Status OK; Proficio USB seen
- [] Optional: **mscc-ui** on Pi, or Windows WPF to Pi IP
- [] Optional: Remote Audio + MscRemotePhones
- [] Optional: Firmware Upload with correct .cyacd